Mohammed Zaeem *M.Sc. Data Science*

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PROFILE SUMMARY

M.Sc. Data Science student at TU Hamburg with hands-on experience in **Python, backend development, data engineering and AI systems**. Experienced in building agentic AI workflows using **tool calling, MCP, retrieval, memory and structured outputs**, as well as developing APIs and data pipelines for reliable technical applications. Comfortable working with **FastAPI, SQL, Git, Docker, cloud platforms and CI/CD**, with a strong interest in LLM evaluation, observability and production-grade AI engineering. Experienced in translating technical requirements into working solutions, testing system behavior and improving reliability through structured experimentation and debugging.

WORK EXPERIENCE

Blade Service Engineer - Working Student, *Nordex* 03/2026 – Present | Hamburg, Germany

- Analyzed engineering and operational data using **SQL, Python and Pandas**, translating technical questions into structured datasets, KPIs and actionable insights for stakeholders.
- Built and maintained **DBT models** for recurring analytics use cases, including data transformations, testing and data-quality validation.
- Worked with technical stakeholders to **define KPIs and measurement approaches**, translating requirements into clear metrics and analytical outputs.
- Developed dashboards and analytical applications using **Tableau, Streamlit and Power BI** to monitor operational performance, trends and anomalies.
- Supported scheduled analytical workflows using **Airflow**, investigated pipeline and data-quality issues, and maintained documentation of metrics, datasets and analytical logic.

Associate || Software Engineer, *Capgemini* 06/2022 – 09/2024

- Developed Python and SQL workflows for data cleaning, transformation, validation and recurring analytical tasks.
- Worked with **Databricks, PySpark and Data Lakes** to process and harmonize datasets from multiple sources.
- Developed backend services and **REST APIs**, connecting applications, databases and data-processing workflows.
- Performed debugging, testing and root-cause analysis and used **Git and CI/CD pipelines** to support reliable software and data workflows.
- Investigated data-quality issues, performed root-cause analysis and validated corrections across data workflows.
- Maintained technical documentation, data-processing logic and change records to support **traceability and reproducibility.**

PROJECTS

Agentic AI Sales Coach (Python, FastAPI, LangGraph, MCP, PostgreSQL, Docker, GCP)

- Built a multi-agent AI system that simulates customer conversations, retrieves relevant context and generates structured coaching feedback using **LangGraph, tool calling, memory and retrieval**.
- Developed a custom **MCP server** that gives agents access to customer profiles, product knowledge, conversation history and evaluation tools through structured interfaces.
- Created an evaluation pipeline using test conversations, **LLM-as-judge, structured scoring and tracing** to compare agent versions and measure response quality, latency and failures.
- Built the backend with **FastAPI**, containerized it with Docker and deployed it on **GCP Cloud Run** with logging, retries and fallback handling for more reliable agent execution.

Radiation Tracking System - Real-Time Big Data Dashboard (Kafka, Flink, PostgreSQL, Dash,
Hamburg University of Technology (6 *ECTS*)

- Developed a scalable, fault-tolerant real-time **data streaming pipeline** using an **ELT approach** with **Python, Apache Kafka** (Confluent)**,** and **PyFlink**.
- Collaborated as part of a **4-member team** to process 29GB+ and billions of SafeCast radiation records in real time.
- Built a containerized system enabling live **CPM** aggregation and continuous data processing.
- Implemented real-time updates using Server-Sent Events (SSE) and deployed the solution on **Google Cloud Platform (GCP)**.

LLM Reasoning & Annotation Analysis Project (Python, PyTorch, Scikit-learn, Transformers)

- Evaluated **zero-shot, few-shot and Chain-of-Thought** prompting for text classification, comparing performance using Accuracy and Macro-F1.
- Built Python-based evaluation workflows to automate prompt experiments and compare AI approaches across multiple test cases.
- Analyzed **GPT-Neo-125M** attention patterns and prediction errors to understand model behavior and identify improvement opportunities.
- Developed a **Tree-of-Thought** AI workflow for generating, evaluating and selecting hypotheses, and summarized results into structured recommendations
- Developed **Python workflows using the OpenAI API** to automate prompt execution, process model responses, enforce structured outputs and evaluate results across multiple experiments.

SEMINAR

Federated Learning under Distributed Concept Drift

- Conducted an in-depth study of **Federated Learning**, focusing on non-IID data, client heterogeneity, privacy-preserving training, and concept drift.
- Analysed and compared **FL+HC, FedDrift, and FairFedDrift** for adaptive model training and group-specific drift detection.
- Evaluated the frameworks' scalability, fairness, computational limitations, and applicability in dynamic real-world environments.

EDUCATION

Masters of Science - Data Science, *Technical university of Hamburg* 10/2024 – 04/2027
Course work: Advanced Machine Learning, Big Data, Statistical Modeling, Artificial Intelligence

Bachelor of Science - Computer Science and Engineering, *Presidency University* 06/2018 – 07/2022
Course work: Artificial Intelligence, Software Engineering and Project Management.

TECHNICAL SKILLS

Agentic AI & LLMs: Python, LangChain, LangGraph, MCP, Tool Calling, RAG, Memory, Context Engineering, Routing, Structured Outputs, Prompt Engineering
LLM Evaluation & Observability: LLM-as-Judge, Evaluation Datasets, A/B Testing, Tracing, Latency & Cost Monitoring, Langfuse
Backend & Data: FastAPI, REST APIs, SQL, PostgreSQL, Async Python, Pandas, NumPy, Scikit-learn
Cloud & DevOps: GCP, Cloud Run, Docker, CI/CD, Git, Logging, Monitoring, Testing, Debugging
Data & Streaming: Apache Kafka, Data Pipelines, Data Validation, ETL, Real-Time Processing
AI Development Tools: ChatGPT Codex, Claude Code, GitHub Copilot

INTERESTS

Hobby
Calisthenics, Badminton, Cooking, Hiking.